

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol

An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence. Where no observation exists, emem signs an absence and its reason, not a zero.

A compact citation keeps the observation traceable across agents and time. Design goal: encode in orbit, decode in AI's reasoning with emem.

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From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor Keep the source scene.	Location · product · time Lookup identity	Canonical record + CID Batch attestation	Pass the reference MCP · REST · A2A	Retrieve the same record Recover its provenance.	Hash · bind · sign · source Policy: refuse if one fails.	Reason with cited evidence Carry the reference onward.

1 Does agent B act on a corrupted handoff?

The same corruptions, five handoff conditions.

Agent A cites a satellite reading. A relay or a faulty signer changes the evidence. Agent B acts on it, declines it or uses the genuine record.

Agent A cites	handoff	relay	Agent B	B acted on corrupted evidence agents, pooled Claude	declined	used the genuine record
 Agent A reads NDVI 0.4709 and cites it Sentinel-2A L2A, 25 Sep 2026 outlined: the 10 m pixel the record names RGB from recorded grids	prose The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Laha...	relay or faulty signer one change per trial, shared corruption families: value · unit · cell · time · band · source · derivation · pixel · stale record · signature / id	B reads the text	254 / 276	22	0
	JSON { "about": { "field": "Keylong, Lahaul (32.57126 N, 77.03448...		B reads the fields	164 / 276	106	6
	retrieved text (RAG) Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell ...		B retrieves a passage	220 / 276	56	0
	opaque id record ref obs-7d1c44		B fetches the sender's record	154 / 288	88	46
	emem reference + verifier emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk6...		B resolve re-hash place / time / band signature log recompute source re-read	0 / 300	264	36

Across the 23 shared items, agents acted on 254 of 276 corrupted handoffs in prose, 154 of 276 with an opaque id, and 0 of 276 with an emem reference (2 of 276 without the instruction to check).

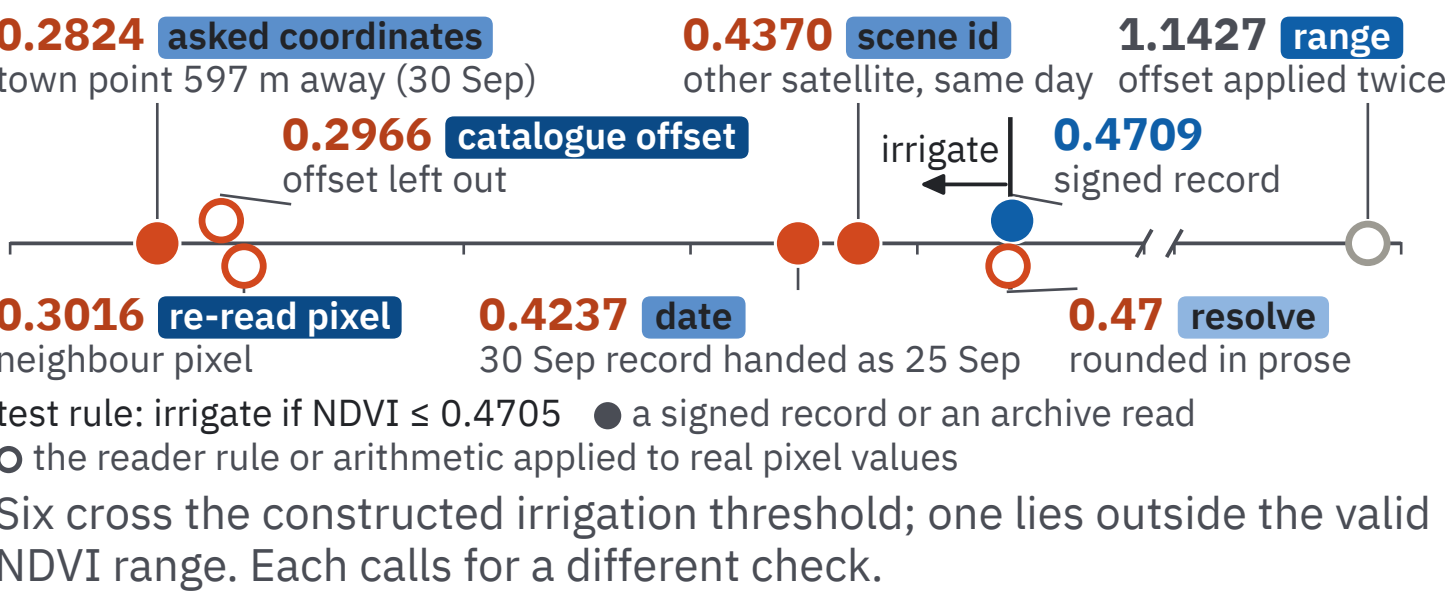
By design, Earth data stays on the encoding device in orbit; its signed record and reference are downlinked, so no source re-read (SAT-042 harness, not flown).

Figure: Haiku 4.5, Sonnet 5.5 and Opus 5.5 pooled, full item set per condition. 2,794 scored trials (1 Oct 2026, two constructed tasks) span 25 corruption items, controls, exploratory items and Qwen. The emem lane adds a verifier and an instruction. Matched baseline (4 Oct): a JSON receiver with emem's checks minus hash, signature and log accepted 8 of 23 (3 with four more checks; emem 0); instructed Haiku agents with it acted on the same 8. With a fail-closed resolver, Qwen2.5-7B still acted on 10 of 25 (2 of 25 instructed): enforce refusal outside the model.

2 How does a value change between agents?

One NDVI, eight values

We varied one Keylong NDVI by scene, date, place, pixel and arithmetic.



3 Why can a value change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

Δz : change in readout. The terms name possible causes (environment, sensor, geolocation, encoder, residual); we have not measured their shares.

7 What does a checked reference establish?

The record, not physical truth

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token **record** **source pixel**

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG[row, col] = DN(b) & Ed25519(a) & leaf in STH & STH_{geo,q} prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.

1-3 content address 1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks... still trusted: BLAKE3 and the verifier's code

4-5 cell binding hashed cell = token cell = the lat/long the derivation used · the cell rule

6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code

8-9 upstream the STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · this STAC item carries no checksum

10-11 server signature emem.dev's Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right

12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the signed tree head (STH) geo.qa co-signed · geo.qa is also run by Vortex AI

15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION

Checkable observation handoff

EO evidence that travels between agents, checked by the receiver down to the source pixel.

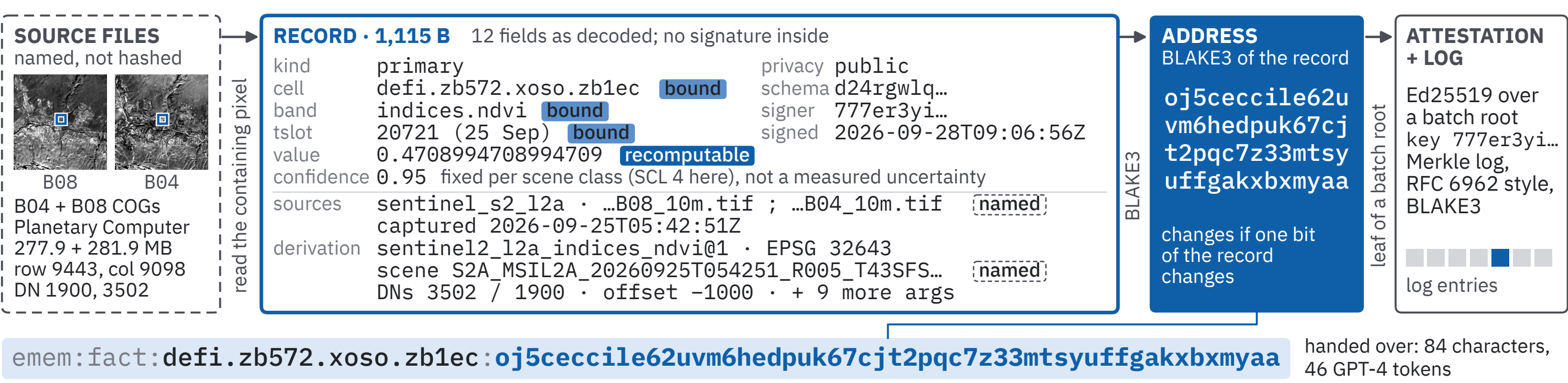
Content hashes (IPFS), signed logs (SCITT) and provenance (PROV) already exist. emem adds typed EO references and a source re-read, which caught a wrong pixel that hashing, signatures, logs and recomputation had passed (panel 5).

4 What exactly is handed over?

A compact reference to an exact record

The receiver uses an 84-character reference to fetch the record and its evidence.

INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.



emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk67cj2ppqcz33mtsuyffgaxkxbmyaa handed over: 84 characters, 46 GPT-4 tokens

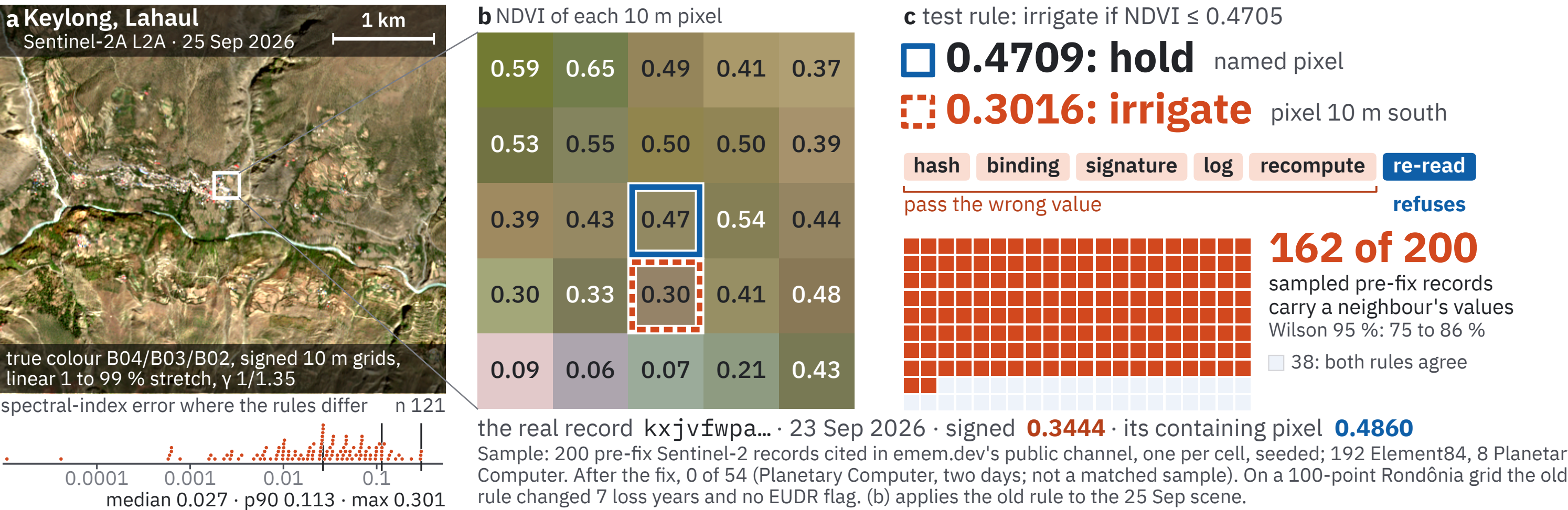
BLAKE3 identifies the 1,115-byte record. Its fields name the source files, sampled point and derivation; a batch signature covers the record address. Source-file hashes are absent from this record.

THE RECORD, AS CODED · fact.rs, cbor.rs, geo.rs

```
fact f = (kind, cell, band, tslot, value, unit, confidence, sources, derivation[fn key, args], privacy, schema_cid, signer, signed_at)
name b = CBOR(f), declaration order, NaN as f9 7e00, -0 as +0; fact_cid = base32(BLAKE3-256(b)); token emem:fact:cell:<fact_cid>
cell lat_q = round((lat+90)/180*(222-1)), lng_q = round((lng+180)/360*(222-1)); pitch 8.583e-5° = 9.52 × 8.06 m here
```

5 A valid record can preserve a wrong source read

NDVI = (DN₈ - DN₄) / (DN₈ + DN₄ + 2o), o = -1000 (product offset)



the real record kxjvfwpa... · 23 Sep 2026 · signed 0.3444 · its containing pixel 0.4860
Sample: 200 pre-fix Sentinel-2 records cited in emem.dev's public channel, one per cell, seeded; 192 Element84, 8 Planetan Computer. After the fix, 0 of 54 Planetary Computer, two days; not a matched sample). On a 100-point Rondônia grid the old rule changed 7 loss years and no EUDR flag. (b) applies the old rule to the 25 Sep scene.

6 Which changes can the receiver detect?

Checks can continue back to the source

Resolve **Re-hash** **Bind** **Signature** **Log** **Recompute** **Re-read**

agent bars: three Claude models pooled, 1 Oct 2026

	prose	JSON	RAG	opaque id	emem checks	first check that refuses
B acted on corrupted evidence	15/15	15/15	15/15	13/16	0/300	0/16
seen in production	15/15	15/15	15/15	13/16	0/300	0/16
value	M1	M2	M3	M4	M5	M6
cell	M7	M8	M9	M10	M11	M12
band	M13	M14	M15	M16	M17	M18
source	M19	M20	M21	M22	M23	M24
derivation	M25	M26	M27	M28	M29	M30
stale / current	M31	M32	M33	M34	M35	M36
signature / id	M37	M38	M39	M40	M41	M42
source pixel	M43	M44	M45	M46	M47	M48
entity	M49	M50	M51	M52	M53	M54

B acted on corrupted evidence

deterministic receiver, same corruptions
resolve hash binding signature log recompute re-read
Bars: agents' false acceptance, 1 of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: the deterministic ceiling. Totals pool 23 to 25 items, 1 Oct 2026. Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Deeper checks expose different corruptions. Agents with an emem reference decided correctly on 71 of 72 genuine controls and declined one.

JSON and an opaque id caught some changes from their fields. On the live pre-fix record, agents acted 4 of 4 times uninstructed, 2 of 4 with resolve-and-bind instructions and 0 of 4 with a source re-read.

8 Can a later agent recover the earlier state?

Revisit the earlier evidence

$$C_t = TxAsOf(O^*, a, b, t)$$

$$\text{recall} = \max_{(t, cid)} \{0 \text{ in } C_t : t \leq t^*\}$$

Latest-as-of mode: take the versions signed by τ (signed_at, the signer's clock), then the latest valid time $\leq t^*$. If no candidate exists, recall returns an empty result; CID order breaks ties.

BENGALURU · SAME CELL AND BAND · 2026

signed 28 May	signed 11 Aug
918.0 m	915.07 m
Open-Meteo / DEM90 earlier CID	Copernicus DEM30 later CID
as of 15 Jun: 918.0 m	

The next agent can recover the earlier citation.

The later 915.07 m reflects a provider change.

9 What does the memory contain?

Observations + temporal edges

$$M = (O^*, E^*) \quad O = (a, b, t, v, u, p, s)$$

a: location cell · b: variable · t: valid time · v: value
u: uncertainty, if recorded · p: provenance, recipe, signed_at
s: attestation associated with the observation

$$E = (\text{subject}, \text{predicate}, \text{object}, t_{\text{from}}, t_{\text{to}})$$

Append observations and supersedes / disagrees_with edges; keep earlier records.

Lookup = (a, b, t); a cid names a record, not an object
cid(record) = BLAKE3(CBOR(record))

Use it in an EO workflow

Compare dates · recall, diff

Assemble evidence · merge, trace

Retain conflicts · competing, evolve

Refresh inputs · ensure, valid

Compare vegetation observations, keep conflicting estimates and pass their references on. A request reuses stored records and fetches and signs missing ones from the archive.

10 What else can an agent carry?

Beyond a single observation

In emem, the token family is how agents read Earth observation: one grammar for a place, a value, a raster, a time series or a device run.

data need	grammar, after emem:	what it id hashes
a place	cell: <cell164>	the quantised lat/long; no hash
one value	fact: <cell>: <cid>	BLAKE3(CBOR(fact)), 52 chars
nothing there	fact: · kind absence	same; + BLAKE3(reason)[:16]
an embedding	fact: · 128 / 1024 floats	same; not re-usable
a derived value	fact: · op, parents	same; pure ops re-usable
a raster	raster: <a>: : <t>: <d>	fact d names BLAKE3(grid)
a raster in time	cube: <a>: : <t0...t1>: <d>	BLAKE3(CBOR([member d...]))
a raster set	rasteriset: <set>: <d>	BLAKE3(CBOR([d..., purpose]))
many facts, one id	bundle: <cid>	BLAKE3(p, cell band t cid)[:16]
a file, one chunk	tree: <file>#row=i	BLAKE3(note)[:16]; root inside
a reasoning stage	state: <cid>	BLAKE3(CBOR(state record))
a named object	entity: <cid>	BLAKE3(anchor)[:16]: a label
a device run	trace: <cid>	BLAKE3(CBOR(signed trace))
an evidence chain	track.v1 note	link = BLAKE3(prev, ref)[:16]

14 data needs; embedding and absence share the fact grammar. Names, sets and records hash differently. Grey: encoders retired; device runs in a reference harness.

One bundle resolves to eight facts; four reasoning-state addresses re-hash. Neither check validates an agent's reasoning.

11 Satellite execution evidence

A signed trace binds the reported run

SAT-042 scripted pass · reference harness · 30 Sep 2026.

Enrol	Write, no trace	Capture the pass	Unbound fact	Bound batch	Score vs anchor	Rewrite one log
ENROLLED	REFUSED	SIGNED	REFUSED	ADMITTED	SCORED	REFUSED
8 trace layers, each log digest chained; seq 2 rewritten after signing; broken at seq 3						
syscall_scheduler memory sensor bus signal energy thermal storage						
seq 0	seq 1	seq 2	seq 3	seq 4	seq 5	seq 6

Reference harness, no spacecraft enrolled. The trace shows which key signed the run and that no logged segment changed; it does not show which code ran on which inputs. The gate binds output digests, not band, cell or time.

12 Use emem. Carry the evidence forward.

ChatGPT · @emem plugin	Claude · Directory plugin / MCP	Diffy · Marketplace plugin	GitHub MCP Registry · Server listing	ClawHub · OpenClaw plugin listing	CONNECT
"Signed facts about real places"	"hand another agent a token it can check offline"	"Shared memory for AI agents. One address per fact, one signature you check."	"Satellites for AI."	"Shared, signed and content-addressed memory of the physical world..."	
Enable emem; paste a token.	Resolve from chat or terminal.	Workflow templates, including EUDR	Install MCP server	Install the OpenClaw plugin.	Setup · EO tasks Reproduce
MORE CLIENTS	PROTOCOLS	DEVELOPER	FRAMEWORK EXAMPLES	DISCOVERY / MIRRORS	
Cursor · VS Code · Gemini CLI	MCP · A2A REST / OpenAPI	Python · TypeScript · Docker LangGraph memory · GitHub source	LangChain · LlamaIndex · AutoGen CrewAI · Mastra · Agno (see setup notes)	MCP Registry · MuleSoft Exchange · Glama · Smithery Context7 · Hugging Face (older mirror)	